

BIMA SALAHUDDIN, IndonesiaWMO: **972700**Lat: **8.5428S**Lon: **118.6930E**Elev: **7**StdP: **14.69**Time Zone: **8.00 (E08)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	68.7	69.9	62.9	86.0	83.4	64.6	91.2	81.3	20.8	85.4	18.9	85.8	0.1	130	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	16.0	95.2	77.8	93.9	77.8	92.7	77.9	82.1	89.7	81.3	88.9	80.7	88.2	11.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	155.8	86.3	79.2	151.7	85.2	78.7	149.0	84.5	45.9	90.0	45.0	89.2	44.3	88.6	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.8	15.2	65.7	98.3	3.4	1.9	63.3	99.6	61.4	100.7	59.5	101.8	57.0	103.2		
(5)			64.1	83.7	2.7	1.2	62.2	84.5	60.6	85.2	59.0	85.8	57.0	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	82.7	82.2	82.5	82.8	82.2	80.5	79.7	80.3	82.6	84.8	85.4	83.6
	DBStd	2.34	1.58	1.47	1.36	1.39	1.59	2.05	1.69	1.73	1.75	1.49	1.88	1.89
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11849	1015	903	1009	985	999	914	922	940	977	1080	1063	1042
	CDD65	6374	550	483	544	535	534	464	457	475	527	615	613	577
	CDH74	65478	5584	4739	5431	5612	5430	4248	3979	4347	5480	7226	7292	6110
	CDH80	25451	1872	1510	1896	2176	2068	1499	1411	1733	2375	3319	3344	2248
(14) Wind	WSAvg	4.4	2.3	2.4	2.5	3.0	4.0	5.2	5.8	6.8	6.8	6.6	4.5	2.6
(15) Precipitation	PrecAvg	53.9	10.5	9.2	7.5	4.3	3.7	1.4	1.2	0.2	0.4	1.5	5.1	9.1
	PrecMax	69.9	17.8	14.8	16.8	11.9	8.2	5.2	6.1	0.7	5.4	6.9	11.6	15.3
	PrecMin	34.9	1.8	4.1	1.7	0.8	2.0	0.7	0.6	0.1	0.0	0.2	0.7	5.1
	PrecStd	9.6	3.5	2.8	3.0	2.8	1.7	1.1	1.4	0.2	1.1	1.6	3.1	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	91.8	92.3	92.9	92.1	90.9	90.1	91.7	94.4	96.8	97.7	95.1
		MCWB	79.6	79.9	79.9	79.7	79.5	78.8	76.7	76.3	75.7	76.4	77.9	79.4
	2%	DB	91.4	90.2	90.9	91.6	90.8	89.4	88.8	90.2	92.8	94.9	95.7	92.7
		MCWB	79.4	79.7	79.6	79.4	78.9	78.0	75.9	75.7	75.9	76.9	78.2	79.9
	5%	DB	89.7	89.0	89.7	90.5	89.8	88.2	87.8	89.1	91.6	93.6	94.2	90.9
		MCWB	79.3	79.5	79.6	79.1	78.6	77.2	75.4	75.3	75.9	77.0	78.3	79.7
	10%	DB	88.1	87.6	88.3	89.4	88.7	87.1	86.7	87.9	90.3	92.4	92.6	89.1
		MCWB	79.1	79.1	79.3	78.9	78.2	76.5	75.0	74.8	75.5	77.0	78.2	79.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.4	82.6	82.7	82.8	82.4	81.1	80.5	80.2	80.5	80.9	82.0	82.4
		MCDB	90.2	88.8	89.5	89.9	89.3	88.8	87.2	88.8	90.2	91.5	92.1	91.1
	2%	WB	81.0	81.0	81.1	81.3	80.8	79.6	78.6	78.0	79.0	79.8	80.8	81.3
		MCDB	88.2	87.6	87.9	88.5	87.8	87.0	85.2	86.7	88.2	90.0	90.0	89.2
	5%	WB	80.1	80.1	80.3	80.3	79.8	78.5	77.4	76.7	77.8	78.9	80.0	80.6
		MCDB	86.8	86.5	87.0	87.4	86.8	85.7	84.4	86.0	86.8	88.9	88.7	88.0
	10%	WB	79.4	79.4	79.6	79.6	79.0	77.6	76.4	75.7	76.8	78.2	79.3	79.9
		MCDB	85.7	85.4	86.1	86.5	86.1	84.5	83.5	85.0	85.6	87.7	87.7	86.7
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	11.8	13.0	14.3	14.4	14.9	15.7	17.3	17.8	17.2	16.0	12.9
		MCDBR	13.9	13.6	14.8	15.9	15.4	15.8	16.6	18.1	18.8	18.4	17.6	14.8
		MCWBR	4.7	5.0	5.4	5.3	5.3	5.9	6.1	6.1	5.1	4.6	4.4	4.6
	5% WB	MCDBR	12.5	12.5	13.6	14.0	13.7	14.1	14.4	17.1	17.0	16.8	15.8	13.6
		MCWBR	5.0	5.2	5.4	5.5	5.3	5.8	6.2	6.7	5.9	5.2	5.0	4.9
(40) Clear-Sky Solar Irradiance	taub	0.451	0.455	0.428	0.425	0.427	0.423	0.419	0.429	0.449	0.484	0.484	0.460	
	taud	2.349	2.335	2.403	2.420	2.412	2.426	2.422	2.374	2.308	2.197	2.229	2.320	
	Ebn at Noon	283	281	284	276	266	262	266	271	274	270	272	279	
	Edh at Noon	42	43	39	37	36	34	35	38	43	49	47	43	
(44) All-Sky Solar Radiation	RadAvg	1683	1701	1769	1764	1661	1560	1616	1801	2017	2073	1975	1721	
	RadStd	168	132	152	116	82	78	59	46	72	94	128	135	

Historical Trends

Station Only	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A